Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	STA2001
Course title (English)	Probability and Statistics I
Course title (Chinese)	概率及统计（一）
Units	3
Language of Instruction	English
Description (English)	This course is to study the basic concepts of probability and statistics. Topics include elementary probability theory, random variables, probability distributions, sampling distributions, convergence of random variables, laws of large numbers and central limit theorem.
Description (Chinese)	本课程的目的是学习概率统计的基本概念。主题包括初等概率论，贝叶斯定理，随机变量，概率分布，数学期望，条件分布，独立性，相关性，一些特殊的单变量和多变量随机分布，随机变量变换，抽样分布，动差生成函数，大数定律和中心极限定理。

2. Prerequisites / Co-requisites

A. Prerequisites

MAT1001 Calculus I, or MAT 1011 Calculus (Extended) I, or MAT1003 Analysis I, or MAT1010 Calculus for Economic Analysis I

Remark: The bivariate continuous random variable part in STA2001 is based on the Multivariable Calculus and therefore, the students are suggested to have some fundamental knowledge of Multivariable Calculus, which can be learned from MAT1002 Calculus II or MAT 1012 Calculus (Extended) II or MAT3010.

B. Co-requisites

Remark: The bivariate continuous random variable part in STA2001 is based on the Multivariable Calculus and therefore, if not learned previously, the students are suggested to take MAT1002 Calculus II or MAT 1012 Calculus (Extended) II or MAT3010, concurrently with STA2001

3. Learning Outcomes

After taking this course, the students are expected

• Knowledge Outcomes:

o Basic knowledge of probability function and counting techniques;

o Basic knowledge of random variables and probability distributions: univariate, bivariate, and multivariate, discrete and continuous;

o Basic knowledge of widely used probability distributions and their corresponding properties, such as Gaussian distribution;

o Basic knowledge of various statistics of probability distributions: mean, variance, moment generating function;

o Basic knowledge of the properties of functions of random variables: Chebyshev's inequality, central limit theorem, law of large numbers and convergence in probability.

• Skills Outcomes:

o Ability to identify the random variable and determine the type of its distribution given necessary information

o Ability to calculate the statistics of interested distributions

o Ability to apply the properties of the widely known distributions and results to derive the corresponding property of the interested distribution

• Value Outcomes:

o Be prepared to acquire necessary and adequate knowledge for further studies of more advanced theory and methods of probability and statistics

4. Course syllabus

•Basic concepts and properties of probability theory: sample space, events, probability, conditional probability, independent events.

•Random variables, probability distributions, mathematical expectation, mean and variance, moments, joint distributions, independent random variables, discrete and continuous random variables, distributions of functions of a random variable.

•Discrete and continuous distributions: Bernoulli, binomial, Poisson, geometric and negative binomial distributions; uniform, exponential, gamma, beta, Weibull, Pareto, normal, chi-square and Student t-distributions.

•Sampling distributions, convergence of random variables, laws of large numbers and central limit theorem, normal approximations.

5. Assessment Scheme

Component/ method	**% weight**
Assignments	20
Mid-Term Exam	30
Final Exam	50

6. Grade descriptor

General

Grade	**Description**
A	- Demonstrate fully understanding of the course materials in all assignments, midterm and final exams, AND - Outstanding ideas are shown in the assignments, midterm or final exams
A-	- Demonstrate fully understanding of the course materials in all assignments, midterm and final exams
B	- Demonstrate fully understanding of the course materials in MOST of the assignments, midterm and final exams
C	- Demonstrate fully understanding of the course materials in SOME parts of the assignments, midterm and final exams
D	- Demonstrate fully understanding of the course materials in LIMITED parts of the assignments, midterm and final exams
F	- Demonstrate fully understanding of the course materials in VERY LIMITED parts of the assignments, midterm and final exams

7. Feedback for evaluation

Formal Course and Teaching Evaluation
Informal feedback and/or teaching assistant
School retreat and program review

8. Reading

Required Reading

1.Probability and Statistical Inference, 978-0-321-92327-1, Hogg, R. V., Tanis, E. A. Zimmerman, D. L, Pearson, 9TH EDITION, 2015, United States

9. Course components

Activity	**Hours/week**
Lecture	3
Tutorial	1

10. Indicative teaching plan

Week	**Content/ topic/ activity**
001	Sections 1.1-1.2: algebra of sets; sample space, events and probability; properties of probability; methods of enumeration: permutation, sampling with and without replacement. combination. Sections 1.3-1.5: conditional probability, multiplication rule; independent events, dependence; Bayes' theorem, posterior probability; law of total probability.
002	Sections 1.3-1.5: conditional probability, multiplication rule; independent events, dependence; Bayes' theorem, posterior probability; law of total probability.
003	Sections 2.1-2.3: discrete and continuous random variables; cumulative distribution function, probability mass and density functions; expected value, variance, standard deviation; moment generating function and moments.
004	Sections 2.3-2.6: discrete distributions: Bernoulli, binomial, geometric, negative binomial and Poisson distributions.
005	Sections 3.1-3.2: continuous distributions: uniform, exponential, gamma, beta distributions.
006	Sections 3.2-3.3: normal distributions
007	Half-time Review and Mid-term Exam
008	Sections 4.1-4.2: joint distributions: joint cumulative distribution, probability mass and density functions, marginal and conditional distributions, covariance and correlation
009	Sections 4.3-4.5: marginal and conditional distributions, covariance, correlation, bivariate distribution of continuous type, bivariate normal distribution
010	Sections 5.1-5.3 transformations of random variables, several random variables, random number generator, histogram
011	Sections 5.3-5.5 several random variables, moment generating function technique, functions of normal random variables, sampling distributions: sample mean, sample variance, and statistics, chi-square distribution, student's t distribution

012	Sections 5.5-5.7 Central limit theorem, approximations for discrete distributions, histogram
013	Section 5.7-5.9 Chebyshev's inequality, convergence in probability, limiting moment generating function technique, law of large numbers.
014	Full-time Review and Q&A

11. Any other information

This course is required for all students in the School of Science and Engineering. It should be taken in the first year of study.

12. Version date

Version number	4
As of (date)	2023/11/22